Learning-Based Trajectory Tracking and Balance Control for Bicycle Robots With a Pendulum: A Gaussian Process Approach

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***Abstract*—This article presents a learning-based control framework for a class of underactuated bicycle robots with an active pendulum attached to the rear body as a balancer. In contrast with the extent methods that require the exact dynamic modeling, the proposed solution deals with the uncertainties, including the impacts of changing road environment, unmodeled dynamics, and external disturbances by virtue of Gaussian process and disturbance cancellation on control device. The tracking controller using traction and steering is presented based on Lyapunov redesign. The self-balancing control of the roll-pendulum underactuated subsystem is considered. This subsystem is first converted into a pure-feedback form, and a novel backstepping controller based on dynamic inversion technique is developed to maintain the roll and pendulum angles in the balance equilibrium manifold. By using an event-triggered update in the learning process, efficient prediction can be achieved, and the tracking error and the roll angle are limited to smaller bounds that are explicitly given. The simulation verifies that adjustable performance can be offered for tracking by tuning the control gains and the hyperparameters of the Gaussian process predictor. Finally, experiments are conducted on a real bicycle platform to illustrate the effectiveness of the control framework.**

***Index Terms*—Backstepping, bicycle robots, learning control, nonlinear underactuated uncertain systems, trajectory tracking.**

I. Introduction

SINGLE-TRACK two-wheel vehicles, such as bicycles, have been a common mean of transportation in human life, due to their flexibility and terrain adaptability [1]. The issue of modeling and controlling autonomous bicycles has, thus, drawn considerable attention in the research intersection of robotics and control. Control of such underactuated robots is intractable since no analytical casual compensator can be derived to attain accurate trajectory tracking while keeping the attitude dynamics stable [2]. In view of this, many works mainly focus on balancing control [3] or straight-line tracking [4].

To simultaneously achieve trajectory tracking and balance for a bicycle, different approaches have been proposed to handle the underactuated characteristic, and can be classified into three groups: control by using steering, flywheel, and pendulum. For the first group, the external/internal control scheme is well developed for trajectory tracking by capturing the balance equilibrium manifold of the roll subsystem [1], [5]. The autonomous bicycle without any balancer may be sensitive to parameter variations and disturbances. If the bicycle is attached to a flywheel, self-balancing can be easily achieved with flywheel torque against the bicycle's falling [6], [7]. In balancing by pendulum, the bicycle with a pendulum can shift its center of gravity corresponding to its attitude so that it can remain (nearly) upright. This approach reflects a more realistic balancing manner of a rider–bicycle system. In [8], a linearized model of a bicycle with a pendulum is constructed and a linear quadratic regulator (LQR) is realized on the system to fulfill balance only. The first systematic investigation of the co-design for balancing and tracking by using pendulum was reported by Keo and Yamakita [9]. However, the linearization around the upright position is unavoidable to get the expression of the posture controller.

Most of the abovementioned control techniques for bicycles rely on exact models of bicycle systems [1], [5]–[7], or can only deal with certain small external disturbances [3], [4]. Indeed, it is almost impossible to obtain an accurate model of a bicycle in complicated, dynamic environments. One solution to cope with imprecise modeling is combining disturbance observers (DO) [10] in the controller design procedure. Similar to the methods in [3] and [4], the DO-based control is established by supposing that the system uncertainties are in a small magnitude and vary slowly. This assumption may be inappropriate in bicycle systems because impacts of bumps and holes on roads can be impulsive, and large uncertainties, such as lateral force, may also exist. These observations drive us to comprehensively address the problem of tracking controller design for pendulum-assist bicycles with full consideration of total uncertainties, including

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unmodeled dynamics, parametric errors, and external disturbances.

On the other hand, data-driven modeling techniques provide efficient tools for system identification and dealing with uncertainties and disturbances in many practical control problems [11]. Among these intelligent tools, Gaussian process (GP), a Bayesian nonparametric regression method, is an effective and relatively convenient way, which has been broadly applied to robot and flight control [12], [13]. Different to many other system identification methods, such as neural networks and fuzzy systems, GP can provide a description of the model fidelity, which is appealing for feedback control design and closed-loop performance analysis. In view of this, the interactions of GP with model predictive control [14] and feedback linearization [15], etc., have been studied in recent literature, respectively. The GP-based control for underactuated robots is reported in [16] and [17]. However, these works do not make full use of the upper bound information of estimation error provided by GP models, so that the control accuracy mostly depends on the identification capability and the tracking errors are limited to fixed bounds. Besides, few results have been published on the GP-based control design for pendulum-assist bicycles.

Based on these observations, we attempt to employ the power of GP regression for the tracking control of a bicycle robot with a pendulum. We first construct a dynamic model for a bicycle robot with a down pendulum by physical and geometric principles. Then, three GP models are established to capture the uncertainties attached in the system separately. Based on the partially data-driven model, the control design is partitioned into external tracking control and internal balancing control. The virtual control signals generated from external are considered as nonautonomous perturbations in the roll-pendulum subsystem. The proposed controller is also applicable to more general underactuated systems [18], such as two-link manipulators. The main contributions of this article lie in the following three aspects.

1) The trajectory tracking control of autonomous bicycles with pendulum via nonlinear approaches is first investigated in this article. This work systematically addresses the effects of system uncertainties by employing online GP learning. The impact of modeling and environmental uncertainties is well accommodated by introducing an "active estimation and compensation" framework.
2) Efficient and computationally inexpensive co-design of learning and control is realized by employing an event-triggered mechanism, which is able to limit the identification errors to prescribed bounds and reduce unnecessary data updates. As a result, sufficiently high tracking and balancing accuracy can be guaranteed by determining the parameters of the GP models and the controllers.
3) For the internal roll-pendulum subsystem (regarded as a two-link system), we propose a novel antidisturbance dynamic inversion (DI) backstepping control scheme to address the mismatched uncertainties. This method, with faster convergence rate and a simpler structure compared to fuzzy or neural backstepping methods [19], can be straightforwardly applied to general pure-feedback systems.

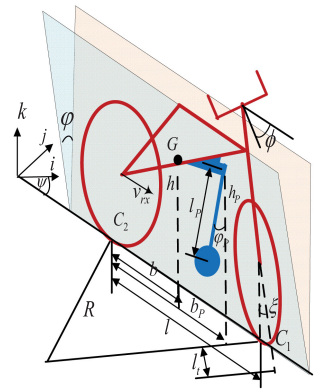

Fig. 1. Notations of the bicycle.

TABLE I
DIAGRAM OF BICYCLE STRUCTURE

Symbols	Meanings
l	Wheelbase
l_t	Trail
G	Mass Center of the bicycle body
P	Pendulum bob
b	Distance from C_2 to projection of G
Ψ	Yaw angle
φ	Roll angle of the bicycle body
b_P	Longitudinal distance from C_1 to pendulum center
l_P	Pendulum length
φ_P	Pendulum's rotation angle
h	Height of the mass center G
h_P	Height of the pendulum's rotation axis
ξ	Caster angle
ϕ	Real steering angle
ϕ_g	Horizontal projection of ϕ
R	Instantaneous turning radius of C_2
m	Mass of the bicycle
m_P	Mass of the pendulum
v_{rx}	Longitudinal velocity of C_2
v_{ry}	Lateral velocity of C_2

A. Notations

The set $\mathbb{R}^+$ is the set of positive reals while the set $\mathbb{R}^{m\times n}$ denotes the set of $m \times n$-dimensional matrices. A^T denotes the transpose of matrix A. The identity matrix with dimension $n \times n$ is written as I_n. $\lambda_{\max}(P)$ and $\lambda_{\min}(P)$ represent the maximum and minimum eigenvalues of a symmetric positive-definite matrix P, respectively. $\Pr\{a\}$ refers to the probability that condition a holds. For convenience, c_θ and s_θ are used to denote $\cos\theta$ and $\sin\theta$, respectively.

II. Modeling of the Bicycle

Based on our previous work on intelligent bicycles [20], a pendulum is added on the rear body (see Fig. 1). The notations in Table I are used in this article.

The geometry, kinematics, and dynamics of the bicycle with a pendulum is systematically analyzed based on the constrained Lagragian method. Then, we can get the model of the bicycle as

$$\begin{cases} \text{Dynamics}: \ M\dot{q} = K + Bu + B_\sigma + D \\ \text{Kinematics}: \dot{p}_x = c_\Psi v_{rx},\ \dot{p}_y = s_\Psi v_{rx},\ \dot{\Psi} = \frac{v_{rx}\sigma}{l} \end{cases} \tag{1}$$

where $q = [\dot{\varphi},\ \dot{\varphi}_P,\ v_{rx}]^{\mathrm{T}} \in \mathbb{R}^3$ is the generalized coordinate vector, $u = [u_\sigma,\ T_P,\ F_{\text{trac}}] \in \mathbb{R}^3$ is the input signal with $u_\sigma = \dot{\sigma}$, T_P, and F_{trac} as the steering input, the pendulum torque input, and the traction force input, respectively, p_x and p_y refer to the

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Cartesian coordinates of the rear wheel contact point C_2, $M \in \mathbb{R}^{3\times3}$, $K \in \mathbb{R}^{3\times1}$, and $B \in \mathbb{R}^{3\times3}$ denote the inertia, Coriolis centrifugal, and potential terms of the bicycle, respectively, and $D \in \mathbb{R}^{3\times1}$ represents the system's uncertainties comprised of some unmodeled dynamics, e.g., lateral movement and external disturbance, such as wind resistance and lateral disturbance force caused by bumps and hollows. In this article, the uncertainties are assumed to be the function of the global positions p_x and p_y and the body state variables φ, φ_P, and q. The structures of the matrices M, K, B, B_σ, and D are specified as

$$M = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}, \quad K = \begin{bmatrix} K_1 \\ K_2 \\ K_3 \end{bmatrix}$$

$$B = \begin{bmatrix} B_1 & 0 & 0 \\ B_2 & 1 & 0 \\ B_3 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} D_1 \\ D_2, \\ D_3 \end{bmatrix}, B_\sigma = \begin{bmatrix} -J_s \frac{c_\varphi s_\varphi}{c_\xi^2} u_\sigma^2 \\ 0 \\ 0 \end{bmatrix} \tag{2}$$

where M_{ij}, K_i, and B_i $(i, j = 1, 2, 3)$ are given in the Supplementary Material, which provides the dynamics analysis and the detailed description of the bicycle model.

If the steering mechanism is supposed to be in the torque mode, its control input is redefined as the steering torque T_S, and the steering dynamics can be specified as $M_4\ddot{\sigma} = K_4 + T_S$.

The control objective of this article is to design viable control laws for the traction, steering torque, and pendulum torque so that the autonomous bicycle can track a reference trajectory within a predetermined error range and keep balance at a safe roll angle. The actuators of the traction and steering will be incorporated for the tracking task, while the stabilization of roll dynamics will be completed through the deflection of the pendulum.

The controller design is based on the model in (1). We design the ideal controller according to the control objectives and the analytical forms of dynamics and kinematics in (1). Meanwhile, considering the deviation of the actual system, we design the disturbance-compensation strategy based on the GP learning algorithm. The proposed tracking and balancing control schemes are illustrated in Fig. 2.

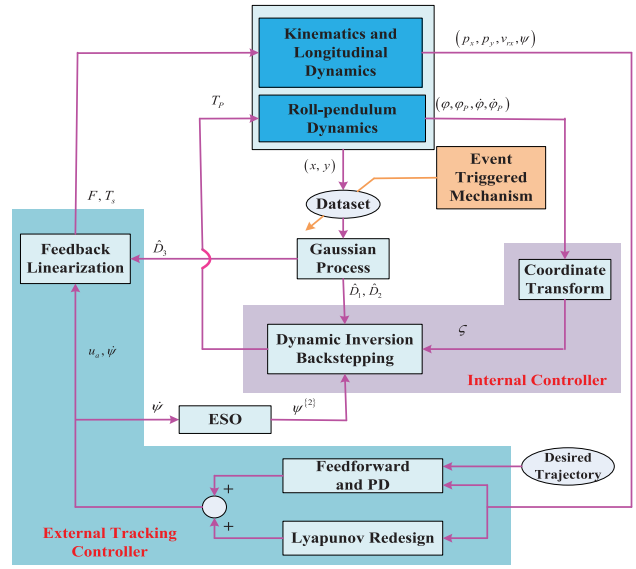

Fig. 2. Proposed control framework.

III. Learning for Identification

For getting the information of the unknown dynamics D in (1), the GP, which can approximate the exact mapping relation of the uncertain function D, is considered. Superior to Kalman filters or extended state observers (ESO) that are widely used as disturbance estimators, GP can estimate complex uncertainties by using limited data, as well as evaluating the modeling fidelity by predicting the upper bound of the estimation error. Besides, GP can be trained online or offline. In this article, we focus on online learning. For this purpose, we construct a dataset $\mathcal{D}_\kappa = \{x^{\{i\}}, y^{\{i\}}\}_{i=1}^{N(\kappa)}$ to store the training input points $x = [p_x,\ p_y,\ \varphi,\ \varphi_P,\ \dot{\varphi},\ \dot{\varphi}_P,\ v_{rx}]^T \in \mathbb{X} \subseteq \mathbb{R}^7$ and output points $y = M\dot{q} - K - Bu - B_\sigma + \Delta \in \mathbb{R}^3$. Here, $\mathbb{X}$ is a compact set, $\Delta \in \mathbb{R}^3$ is the independent and identically distributed measurement noise, which is assumed to satisfy $\Delta_i \sim \mathcal{N}(0,\ \sigma_{\text{gp}}^2)$, $\text{i} = 1,\ 2,\ 3$, t_κ is the update time when a new training data point arrives in the dataset, and $N(\kappa)$ refers to the number of the training data points during the time interval $[t_\kappa,\ t_{\kappa+1})$. The training data in the dataset can change over time, which means that the learning algorithm allows online learning. Since the bumps and slopes on the ground may have effects on the bicycle dynamics, which are related to the position, the variables p_x and p_y are included in the training dataset.

By defining the input matrix $X = [x^{\{1\}},\ x^{\{2\}} \ldots,\ x^{\{N_\kappa\}}] \in \mathbb{R}^{7\times N_\kappa}$ and the output matrix $Y = [y^{\{1\}},\ y^{\{2\}} \ldots,\ y^{\{N_\kappa\}}] \in \mathbb{R}^{3\times N_\kappa}$. The prediction of the uncertainties D at a test data point $x^+ \in \mathbb{X}$ is provided by

$$\hat{D}_i\left(x^+|\mathcal{D}_\kappa\right) = m_i\left(x^+\right) + \mathbf{k}(x^+,\ X)^T\left(K_k + \sigma_{gp}^2 I_{N_\kappa}\right)^{-1} \left(Y_i - \left[m_i\left(x^{\{1\}}\right),\ \ldots,\ m_i\left(x^{\{N_\kappa\}}\right)\right]\right) \tag{3}$$

$$v_i\left(x^+|\mathcal{D}_\kappa\right) = k\left(x^+, x^+\right) - \mathbf{k}(x^+, X)^T \left(K_k + \sigma_{gp}^2 I_{N_\kappa}\right)^{-1}\mathbf{k}(x^+, X) \tag{4}$$

for $i = 1,\ 2,\ 3$. In (3) and (4), Y_i denotes the ith column of Y, $\hat{D}_i(x^+|\mathcal{D}_\kappa)$ and $v_i(x^+|\mathcal{D}_\kappa)$ are the posterior mean and variance function, respectively, $m_i(\cdot)\ :\ \mathbb{X} \to \mathbb{R}$ is the mean function, $k(\cdot):\ \mathbb{X} \times \mathbb{X} \to \mathbb{R}$ is the kernel, and

$$K_k = \begin{bmatrix} k\left(x^{\{1\}}, x^{\{1\}}\right) & \cdots & k\left(x^{\{1\}}, x^{\{N_\kappa\}}\right) \\ \vdots & \ddots & \vdots \\ k\left(x^{\{N_\kappa\}}, x^{\{1\}}\right) & \cdots & k\left(x^{\{N_\kappa\}}, x^{\{N_\kappa\}}\right) \end{bmatrix} \in \mathbb{R}^{N_\kappa \times N_\kappa} \tag{5}$$

$$\mathbf{k}\left(x^+,\ X\right) = \left[k\left(x^+,\ x^{\{1\}}\right),\ \ldots,\ k\left(x^+,\ x^{\{N_\kappa\}}\right)\right]^T \in \mathbb{R}^{N_\kappa}. \tag{6}$$

There are two kinds of functions to be designed in GP: the mean function $m_i(\cdot)$ and the kernel function $k(\cdot)$. The mean function incorporates prior knowledge about the uncertainty,

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which can be derived by using other system identification methods in advance. In this article, however, as we have no information about the unknown dynamics, the mean function is taken as zero. The kernel function represents a set of basis feature function in regression problem. One dependable kernel for GP regression is the so-called squared exponential kernel, which is specified as $k(x, x') = \sigma_k^2 \exp(\sum_{i=1}^{7} \frac{\Delta_{xi}^2}{-2l_i^2})$, where $\Delta_{xi} = x_i - x'_i, \sigma_k, l_i \in \mathbb{R}^+$ are hyperparameters with the meanings of signal variance and length scale, respectively.

The GP model with squared exponential kernels has the capability of universal approximation and, thus, can realize exact modeling of any smooth function. The following lemma describes the approximation error bound in the probabilistic manner.

Lemma 1 ([21]): Let $\mathbb{X}$ be a compact set. Assume that the function D has a bounded reproducing kernel Hilbert space norm on $\mathbb{X}$ with respect to the kennel $k(x, x')$, i.e., $\|D_i\|_k < \infty$. Then, for $\delta \in (0,\ 1)$, consider the GP model subject to (3) and (4), the approximation error is bounded by

$$\Pr\left\{\left\|\hat{D}_i\left(x^+|\mathcal{D}_\kappa\right) - D_i\right\| \leqslant \beta_i\sqrt{v_i\left(x^+|\mathcal{D}_\kappa\right)}\right\} \geqslant 1-\delta \quad \forall x \in \mathbb{X} \tag{7}$$

where $\beta_i = \sqrt{2\|D_i\|_k^2 + 300\gamma_i \ln^3(\frac{N_\kappa+1}{\delta})}$, and γ_i is the maximum information gain. ■

IV. Controller Design and Stability Analysis

The control scheme is divided into two parts. First, the bicycle has to track any smooth desired path trajectory. An external tracking controller is designed based on the feedforward potential difference (PD) and Lyapunov redesign technique. The external virtual control signals will be regarded as perturbations acting on the internal loop dynamics. For the internal dynamics, we expect the bicycle robot to keep balanced in the complicated environment, in which the bicycle may face the effects of lateral interference force, sloping road, or even road/tire friction that cannot be completely modeled, for example. In this case, an internal balancing controller is designed, based on backstepping procedure. Note that both the internal and the external controllers are partly based on the data-driven models from GP, which contributes to the disturbance rejection to some extent.

A. External Tracking Control

In this article, we assume that the desired trajectory $(p_{xd}(t),\ p_{yd}(t))$ satisfies the following assumption.

Assumption 1: $(p_{xd}(t),\ p_{yd}(t))$ and its first and second time derivatives are available and bounded. What is more, $|\dot{p}_{xd}(t)| + |\dot{p}_{yd}(t)| \neq 0$ for all $t \geq 0$.

Remark 1: Assumption 1 is reasonable and can be satisfied in applications since most continuous-curvature-based planning methods can guarantee the smoothness of the reference trajectory, and the derivative of the trajectories can be readily obtained. Inequality $|\dot{p}_{xd}(t)| + |\dot{p}_{yd}(t)| \neq 0$ is the mathematical expression that the bicycle will not stop. Besides, Assumption 1 is pretty common in the existing literature [1], [3], [9], [22] for the trajectory tracking control of bicycles.

With the involvement of GP regression, the model of the bicycle (1) can be rewritten as

$$M\dot{q} = K + Bu + B_\sigma + \hat{D}\left(x|\mathcal{D}_\kappa\right) + e_D \tag{8}$$

where $e_D \triangleq D - \hat{D}(x|\mathcal{D}_\kappa) = [e_{D1},\ e_{D2},\ e_{D3}]^T$ denotes the approximation error of GP.

First, similar to [1], by input transformation $F_{\text{trac}} = M_{31}\ddot{\varphi} + M_{32}\ddot{\varphi}_P - K_3 - B_3\dot{\sigma} - \hat{D}_3(x|\mathcal{D}_\kappa) + M_{33}u_a$, the third equation of (8) becomes

$$\dot{v}_{rx} = u_a + M_{33}^{-1}e_{D3} \tag{9}$$

where u_a is an intermediate control variable to be designed later. In (9), it can be seen that the uncertainty occurred in the third subsystem of (1) will have influence on the tracking performance due to the feedback linearization of F_{trac}.

Differentiate the kinematic model of the bicycle with respect to time and take (9) into account, then we have

$$\begin{bmatrix}\ddot{p}_x\\ \ddot{p}_y\end{bmatrix} = \begin{bmatrix}c_\Psi & -s_\Psi v_{rx}\\ s_\Psi & c_\Psi v_{rx}\end{bmatrix}\begin{bmatrix}u_a + M_{33}^{-1}e_{D3}\\ u_\Psi\end{bmatrix} \tag{10}$$

where $u_\Psi \triangleq \dot{\Psi}{=}\sigma v_{rx}/l$ is another intermediate control variable. Equation (10) is in the feedback-linearizable form without considering the perturbed item $M_{33}^{-1}e_{D3}$. We can thereby design a tracking controller to make the bicycle follow the trajectory $(p_{xd}(t),\ p_{yd}(t))$

$$\begin{bmatrix}u_a\\ u_\Psi\end{bmatrix} = \begin{bmatrix}c_\Psi & s_\Psi\\ \frac{-s_\Psi}{v_{rx}} & \frac{c_\Psi}{v_{rx}}\end{bmatrix}\left(\begin{bmatrix}p_{xd}^{(2)}\\ p_{yd}^{(2)}\end{bmatrix} - \sum_{i=0}^{1} p_i\begin{bmatrix}p_x^{(i)} - p_{xd}^{(i)}\\ p_y^{(i)} - p_{yd}^{(i)}\end{bmatrix} + \begin{bmatrix}u_{rx}\\ u_{ry}\end{bmatrix}\right) \tag{11}$$

where the gain coefficients $p_1,\ p_0$ are selected such that the polynomial $s^2 + p_1 s + p_0$ is Hurwitz, and u_{rx} and u_{ry} are Lyapunov redesign controllers to be determined later.

By substituting (11) into (10) and defining the tracking errors as $e_x = [p_x - p_{xd},\ \dot{p}_x - \dot{p}_{xd}]^T$ and $e_y = [p_y - p_{yd},\ \dot{p}_y - \dot{p}_{yd}]^T$, the tracking error dynamics is obtained as

$$\begin{cases}\dot{e}_x = A_p e_x + B_p\left(c_\Psi M_{33}^{-1}e_{D3} + u_{rx}\right)\\ \dot{e}_y = A_p e_y + B_p\left(s_\Psi M_{33}^{-1}e_{D3} + u_{ry}\right)\end{cases} \tag{12}$$

with $A_p = \begin{bmatrix}0 & 1\\ -p_0 & -p_1\end{bmatrix}$ and $B_p = [0\ \ 1]^T$. According to [23], let P be the solution of the equation $A_p^T P + PA_p = -I_2$. By defining two Lyapunov candidate functions $V_{px} = e_x^T P e_x$ and $V_{py} = e_y^T P e_y$, we obtain the Lyapunov redesign terms u_{rx} and u_{ry} as

$$u_{rx} = -\frac{\partial V_{px}}{\partial e_x}B_p\alpha_3,\ u_{ry} = -\frac{\partial V_{py}}{\partial e_y}B_p\alpha_3 \tag{13}$$

where α_3 is a designed parameter.

So far, we have not specified when new training data points for the GP models are added. Usually, the GP model is updated after a certain fixed interval, i.e., $t_{\kappa+1} = t_\kappa + \Delta_t$ with a positive constant Δ_t. In this article, to achieve sufficient learning, we draw on the idea of [15] in which an event-triggered learning algorithm is employed for system modeling. Specifically, to

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render a balance between the computational complexity and tracking accuracy, we only need to add training data when the approximation error e_{D3} becomes too large. Based on this intuition, the update time of the third GP model [$i = 3$ in (3) and (4)] is designed as

$$t_{3,\,\kappa+1} := \left\{ t > t_{3,\,\kappa} \mid \left|M_{33}^{-1}\right| \beta_3 \sqrt{v_3\left(x|\mathcal{D}_\kappa\right)} > \alpha_3 \right\}. \quad (14)$$

Theorem 1: Consider the kinematic system (10) of the bicycle controlled by (11) and (13). Use the GP model to identify the unknown dynamics D, and the triggered time is designed as (14). Suppose that $\|D_i\|_k < \infty$ for $x \in \mathbb{X}$. If the desired trajectory satisfies Assumption 1, then the control laws (11) and (13) guarantee that the signals $v_{rx},\ u_a,$ and u_Ψ are all bounded, and the tracking error $(e_x,\ e_y)$ is uniformly ultimately bounded (UUB) in probability by

$$\Pr\left\{ \max\left\{\|e_x\|, \|e_y\|\right\} \leqslant \frac{\sqrt{\alpha_3}}{2} \quad \forall t \in [0,\ \infty) \right\} \geqslant 1 - \delta \quad (15)$$

for $x \in \mathbb{X}$. ■

Proof: Please see Appendix A. ■

Remark 2: Theorem 1 characterizes the fundamental tradeoff between the tracking performance and the computational complexity of the GP model. In particular, if the control gain α_3 in (13) is chosen to be larger, it results in lower frequency of adding training data according to (14), and consequently, fewer learning processes over the whole time horizon. A larger value of α_3, on the other hand, also leads to lower accuracy of tracking, whereas a smaller α_3 will contribute to better tracking performance at the cost of more dataset updates.

B. Backstepping Control for Balancing

After applying the tracking control, the original system (1) can be considered as an underactuated two-link system [9] with external perturbations $D_1,\ D_2,\ \sigma,\ \dot{\sigma},$ and $\ddot{\Psi}$

$$\begin{bmatrix} M_{11} & M_{12} \\ M_{12} & M_{22} \end{bmatrix} \begin{bmatrix} \ddot{\varphi} \\ \ddot{\varphi}_P \end{bmatrix} = \begin{bmatrix} K_1 + D_1 - \frac{J_s c_\varphi s_\varphi}{c_\xi^2}\dot{\sigma}^2 \\ K_2 + D_2 \end{bmatrix} + \begin{bmatrix} M_{\Psi_1} \\ M_{\Psi_2} \end{bmatrix} \ddot{\Psi} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} T_P \quad (16)$$

where $M_{\Psi_1} = -M_{13} l/\sigma$, $M_{\Psi_2} = -M_{23} l/\sigma$.

The $\varphi - \varphi_P$ system (16) is treated as an underactuated internal subsystem of (1). The control of (16) meets the challenges in the aspect that the perturbed term $\begin{bmatrix} K_1 + D_1 - J_s c_\varphi s_\varphi \dot{\sigma}^2 / c_\xi^2 \\ K_2 + D_2 \end{bmatrix} + \begin{bmatrix} M_{\Psi_1} \\ M_{\Psi_2} \end{bmatrix} \ddot{\Psi}$ is nonvanishing with respect to the attitude angles. This property is caused by Coriolis when the bicycle is moving and turning around, and will make the equilibrium of the system away from the origin. Some common techniques, such as partial feedback linearization [24] and sliding mode control [18], [25], cannot handle this challenge since they all require that the external disturbances is Lipschitz bounded, and the upright position should be the equilibrium point. These conditions are not satisfied in (16). In this regard, we propose a novel backstepping control scheme to stabilize the internal roll dynamics while remaining the pendulum angle in an acceptable bounded range.

Following the idea in [18], we choose new states and the global transformation of the coordinates by $\varsigma_1 = \varphi + \gamma(\varphi_P)$, $\varsigma_2 = M_{11}\dot{\varphi} + M_{12}\dot{\varphi}_P$, $\varsigma_3 = \varphi_P$, and $\varsigma_4 = \dot{\varphi}_P$, where $\gamma(\varphi_P) = \int_0^{\varphi_P} M_{11}^{-1}(s) M_{12}(s) ds$. By change of coordinates, the system (16) is converted into a nonlinear pure feedback form

$$\begin{cases} \dot{\varsigma}_1 = M_{11}^{-1}\varsigma_2 \\ \dot{\varsigma}_2 = f_1(\varsigma_1,\ \varsigma_3) + g_1\left(\varphi,\ \sigma,\ \dot{\sigma},\ v_{rx},\ \ddot{\Psi}\right) + D_1 \\ \dot{\varsigma}_3 = \varsigma_4 \\ \varsigma_4 = f_2(\varsigma) + g_2\left(\varphi,\ \sigma,\ v_{rx},\ \ddot{\Psi}\right) + b_2 T_P + b_1 D_1 + b_2 D_2 \end{cases} \quad (17)$$

where

$$f_1 = mghs_\varphi + m_P g h_P s_\varphi - m_P g l_P s_{\varphi+\varphi_P}$$

$$g_1 = \frac{mgbl_t c_\xi}{l} c_\varphi \sigma + \frac{m_P g b_P l_t c_\xi}{l} c_\varphi \sigma - mh\left(1 - \frac{hs_\varphi \sigma}{l}\right)\frac{c_\varphi \sigma}{l} v_{rx}^2 - m_P h_P \left(1 - \frac{h_P s_\varphi \sigma}{l}\right)\frac{\sigma c_\varphi}{l} v_{rx}^2 - \frac{J_s c_\varphi s_\varphi \dot{\sigma}^2}{c_\xi^2} + M_{\Psi_1}\ddot{\Psi}$$

$$f_2 = b_1(K_1 - g_1) + b_2 K_2,\ g_2 = b_1\left(g_1 + M_{\Psi 1}\ddot{\Psi}\right) + b_2 M_{\Psi 2}\ddot{\Psi}$$

$$b_1 = \frac{-M_{12}}{M_{11}M_{22} - M_{12}^2},\ b_2 = \frac{M_{11}}{M_{11}M_{22} - M_{12}^2}. \quad (18)$$

The control objective for (17) is to make $\varsigma_1 = 0$. Indeed, ς_2 is the generalized momentum conjugate to φ [26], and ς_1 is the integral of the normalized momentum $M_{11}^{-1}\varsigma_2$. Then, the $(\varsigma_1,\ \varsigma_2)$-subsystem of (17) is thereby the Euler–Lagrange equation with configuration vector $(\varsigma_1,\ \varsigma_2)$. The control objective is for the stabilization of this new transformed Lagrangian system at the equilibrium point $(\varsigma_1,\ \varsigma_2) = (0,\ 0)$.

Define three compact sets $\mathbb{B}_\varphi = \{\varphi :\ \varphi \in [-\varphi_{\max},\ \varphi_{\max}]\}$, $\mathbb{B}_{\varphi P} = \{\varphi_P :\ \varphi_P \in [\varphi_{\max} - \frac{\pi}{2},\ \frac{\pi}{2} - \varphi_{\max}]\}$, and $\mathbb{B}_\varsigma = \{\varsigma_1 : \varsigma_1 \in [-\varphi_{\max} - \gamma(\varphi_{\max} - \frac{\pi}{2}),\quad \varphi_{\max} + \gamma(\frac{\pi}{2} - \varphi_{\max})]\}$. Here, $\varphi_{\max}$ denotes the maximum roll angle allowed for a given bicycle [22] for the consideration of lateral friction forces that are not modeled in this article. Usually $\varphi_{\max} > 30°$. $\varphi_{\max}$ can be estimated by $\varphi_{\max} = F_{y,\max}/k_\varphi$, where $F_{y,\max}$ is the maximum lateral force and k_φ refers to the tire camber stiffness coefficient.

It can be checked that system (17) satisfies the following properties.

Property 1: $\frac{\partial f_1}{\partial \varsigma_3} < 0$ for all $(\varphi,\ \varphi_P) \in \mathbb{B}_\varphi \times \mathbb{B}_{\varphi P}$.

Property 2: $M_{11}^{-1} > 0$ and $b_2 > 0$ for all $\varphi_P \in (-\frac{\pi}{2},\ \frac{\pi}{2})$.

Property 3: For any $t \in [0,\ \infty)$, the following inequalities hold:

$$\left|M_{11}^{-1}\right| \leqslant 1/\left(J_B + J_P + mh^2 + m_P(l_P - h_P)^2\right) \triangleq c_1$$

$$\left|\frac{\partial f_1}{\partial \varsigma_3}\right| \leqslant \frac{mgh + m_P g h_P}{2} + m_P g l_P \triangleq c_2$$

with known positive constants c_1 and c_2.

Properties 1 and 2 characterize necessary local-controllability conditions [27] for system (17). Particularly, Properties 1 and 2

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mean that all the (virtual) control signals $\varsigma_2,\ \varsigma_3,$ and u always have influences on the system dynamic response. Remind that Properties 1–3 are all satisfied on the dynamic modeling (1) under the parameter setting given in Table II in the supplementary material.

In view of the abovementioned properties and nonstrict feedback form of (17), in this section, we will present a DI backstepping controller for the underactuated two-link system. Define the scaled error $z_i = \varsigma_i - \eta_{i-1},\ i = 1,\ 2,\ 3,\ 4$, where η_{i-1} are the virtual control signals and $\eta_0 = 0$.

Step 1: Differentiating z_1 with respect to time and substituting the first subsystem of (17) follow that $\dot{z}_1 = M_{11}^{-1}\varsigma_2$. Therefore, the first virtual controller is designed as

$$\varepsilon\dot{\eta}_1 + \eta_1 = \eta_1^* = -k_1 M_{11} z_1 \tag{19}$$

where $\varepsilon \in (0,\ 1)$ and $k_1 > 0$ are constant parameters. The left-hand side of (19) is a filtered version of the virtual control, and is introduced to avoid the repeated differentiations of the virtual control variables.

Step 2: We calculate the time derivative of z_2 along the trajectory of ς_2 and obtain

$$\begin{aligned}\dot{z}_2 &= f_1(\varsigma_1,\ \varsigma_3) + g_1\left(\varphi,\ \sigma,\ \dot{\sigma},\ v_{rx},\ \ddot{\Psi}\right) + D_1 - \dot{\eta}_1 \\ &= \left(f_1(\varsigma_1,\varsigma_3) + g_1\left(\varphi,\sigma,\dot{\sigma},v_{rx},\ddot{\Psi}\right) + D_1 - \dot{\eta}_1 + k_2 z_2\right) - k_2 z_2\end{aligned} \tag{20}$$

where $k_2 > 0$. Let $\eta_2^* \in \mathbb{B}_{\varphi P}$ be the (unknown) solution of ς_3 for the equation $f_1(\varsigma_1,\ \varsigma_3) + g_1(\varphi,\ \sigma,\ \dot{\sigma},\ v_{rx},\ \ddot{\Psi}) + D_1 - \dot{\eta}_1 + k_2 z_2 = 0$, which is the so-called balance equilibrium manifold [1]. The condition $\eta_2^* \in \mathbb{B}_{\varphi P}$ can be achieved by adjusting the length and mass of the pendulum. Thus, according to [28], the following DI technique can be exploited to generate the virtual control η_2 and make η_2 close to η_2^*

$$\text{DI}: \varepsilon\dot{\eta}_2 = f_1(\varsigma_1,\ \eta_2) + g_1\left(\varphi,\sigma,\dot{\sigma},v_{rx},\Psi^{\{2\}}\right) + \hat{D}_1 - \dot{\eta}_1 + k_2 z_2. \tag{21}$$

$\hat{D}_1$, updated by the GP model, is the estimate of the external disturbance D_1. $\Psi^{(2)}$ is the estimate of $\ddot{\Psi}$ and is introduced for the consideration that $\ddot{\Psi}$ cannot be obtained explicitly from the external tracking controller. Here, the following ESO [29], [30] is designed to estimate $\ddot{\Psi}$:

$$\text{ESO}: \begin{cases} \dot{\Psi}^{(1)} = \Psi^{(2)} + l_1\left(u_\Psi - \Psi^{(1)}\right)/\varepsilon^2 \\ \dot{\Psi}^{(2)} = l_2\left(u_\Psi - \Psi^{(1)}\right)/\varepsilon^4 \end{cases} \tag{22}$$

where l_1 and l_2 are the ESO gains selected such that the polynomial $s^2 + l_1 s + l_2$ is Hurwitz.

Since η_2 is presented directly in the form of differential, no filter is needed to process the virtual signal.

Step 3: Similar to Step 1, one can readily get the differentiation of z_3 as $\dot{z}_3 = \varsigma_4 - \dot{\eta}_2$, and the third virtual controller can, thus, be chosen as ς_4 and specified as

$$\varepsilon\dot{\eta}_3 + \eta_3 = \eta_3^* = \dot{\eta}_2 - k_3 z_3 \tag{23}$$

with $k_3 > 0$.

Step 4: In the final step, we take account for the last subsystem of (17). The time derivative of z_4 is f_2+$g_2 + b_2 T_P - \dot{\eta}_3$. The actual control signal, i.e., the pendulum torque is, then, designed as

$$T_P = \frac{-f_2(\varsigma) - g_2\left(\varphi,\sigma,v_{rx},\Psi^{(2)}\right) - b_1\hat{D}_1 - b_2\hat{D}_2 + \dot{\eta}_3 - k_4 z_4}{b_2} \tag{24}$$

where $k_4 > 0$.

Similar to (14), the update time for new training data in the first and second GP models [$i = 1,\ 2$ in (3) and (4)] is event triggered, determined by the approximation errors of D_1 and D_2. To make the effects of the approximation errors e_{D1} and e_{D2} on the system (17) as small as possible, the event-triggered mechanism is designed by

$$\begin{cases} t_{1,\,\kappa+1} = \left\{t > t_{1,\,\kappa} \mid\ \beta_1\sqrt{v_1} > \alpha_1\varepsilon\right\} \\ t_{2,\,\kappa+1} = \left\{t > t_{2,\,\kappa} \mid\ b_1\beta_1\sqrt{v_1} + b_2\beta_2\sqrt{v_2} > \alpha_2\right\} \end{cases} \tag{25}$$

where v_i refers to $v_i(x^+|\mathcal{D}_\kappa)$ in (4), and α_1 and α_2 are hyperparameters.

Combining (17)–(24), we get the closed-loop error dynamics

$$\begin{aligned}\dot{z} &= f_z\left(z,\eta,\Psi^{(2)},\varepsilon\right) \\ \text{with } z &= [z_1,z_2,z_3,z_4]^T,\ \eta = [\eta_1,\eta_2,\eta_3]^T\end{aligned} \tag{26}$$

where $f_z\left(z,\eta,\Psi^{(2)},\varepsilon\right)$ is shown in (27) at the bottom of this page.

The following theorem summarizes the main result of the proposed backstepping balancing controller.

Theorem 2: Consider the transformed system (17) in which the pendulum torque is generated by DI backstepping procedure (19)–(24) and the uncertainties are estimated by the GP models in (3). Suppose that $\|D_i\|_k < \infty$ for $x \in \mathbb{X}$ and the desired trajectory satisfies Assumption 1. If the control gains $k_i,\ i = 1,\ 2,\ 3,\ 4$ are chosen to satisfy $k_1 > c_1/2,\ k_2 > (1 + c_1 + c_2)/2,\ k_3 > (1 + c_2)/2,$ and $k_4 > 1$. Then, there exists a suitably small constant $\varepsilon_1 \in (0,\ 1)$ such that for any $\varepsilon \in (0,\ \varepsilon_1)$, the closed-loop error dynamics (26) achieves UUB results with probability $1 - \delta$ in the sense that for all $z_1(0) \in \mathbb{B}_\varsigma,\ z_2(0) \in \mathbb{R},\ \varphi_P(0)$ and $\varphi_P(0) - z_3(0) \in \mathbb{B}_{\varphi P}$, and $z_4(0) \in \mathbb{R},\ z(t)$ converges exponentially to an ultimate bound that can be made arbitrarily small by appropriately selecting parameters

$$f_z\left(z,\eta,\Psi^{(2)},\varepsilon\right) = \left\{\begin{array}{l} -k_1 z_1 + M_{11}^{-1} z_2 - M_{11}^{-1}\varepsilon\dot{\eta}_1 \\ -k_2 z_2 + \varepsilon\dot{\eta}_2 + f_1(\varsigma_1,\varsigma_3) - f_1(\varsigma_1,\eta_2) + e_{D1} + M_{\Psi 1}\left(\ddot{\Psi} - \Psi^{(2)}\right) \\ -k_3 z_3 + z_4 - \varepsilon\dot{\eta}_3 \\ -k_4 z_4 + b_2 M_{\Psi 2}\left(\ddot{\Psi} - \Psi^{(2)}\right) + b_1 e_{D1} + b_2 e_{D2} \end{array}\right\}. \tag{27}$$

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k_i, ε, and α_2, that is

$$\Pr\left\{\lim_{t\to\infty}\|z(t)\| \leqslant \alpha_2/\sqrt{\lambda_z} + N_\varepsilon\varepsilon\right\} \geqslant 1-\delta \tag{28}$$

with positive constants λ_z and N_ε. ■

Proof: Please see Appendix B. ■

According to Theorem 2, $\varphi_P = \varsigma_3 = z_3 + \eta_2$ is bounded in $\mathbb{B}_{\varphi P}$ and will converge to the small region containing the manifold η_2^*, while $\varsigma_1 = z_1$ is bounded in $\mathbb{B}_\varsigma$ and will converge to the small region containing zero. As a result, the roll angle $\varphi = \varsigma_1 - \gamma(\varphi_P)$ is bounded in $\mathbb{B}_\varphi$, and will certainly stabilize at the region containing $-\gamma(\eta_2^*)$. What is more, the abovementioned regions can be arbitrarily small by adjusting the control parameters k_i, ε, and α_2.

Remark 3: In the control scheme, the impact of modeling and environmental uncertainties is well accommodated by introducing an "active estimation and compensation" framework. Actually, most existing research works on bicycle tracking control [4], [6] did not consider the effects of system uncertainties, or only addressed small affine perturbations [1], [7] and then design robust controllers. Such robustness-based design may lead to conservative choices of control parameters and cannot be applied to systems with large uncertainties. On the other hand, compared with [15], we update the training data of GP according to the upper bound of the prediction error, rather than the temporal derivatives of some Lyapunov functions. With this design, the tracking error can be maintained in a small range clearly determined by the value of α_3 in (15).

Remark 4: The proposed DI backstepping method can be straightforwardly applied to general pure-feedback systems. The existing literature often uses fuzzy or neural networks to construct backstepping frameworks [19] for these systems. These methods have heavy computational burdens since they need to initialize and update many adaptive parameters, whereas the DI has properties of faster convergence and simpler structure.

V. Simulation

In this section, a numerical example is used to demonstrate the control system performance. The physical parameters of the bicycle are given in Table II in the supplementary material.

In the example, the desired trajectory is a shape of "8," comprised of two circles with a radius equal to 2.5 m. The initial states of the bicycle are set as $p_x(0) = 0$, $p_y(0) = 1$, and $v_{rx}(0) = 1.5$. The unknown dynamics $D = [0.1p_x\cos(\varphi) + 0.5,\quad 0.2\varphi^2,\ 0.1v_{rx}^2 + 0.2p_x]^T$ is assumed to act on the system. The values of the control parameters are designed as $\varepsilon = 0.1$, $k_1 = k_2 = k_3 = k_4 = k_5 = 2$, $p_0 = p_1 = 0.5$, and $l_1 = l_2 = 1$. Three GP models defined by (3) are used as oracles to estimate the effects of D with $\alpha_1 = 1$, $\alpha_2 = 0.04$, and $\alpha_3 = 0.02$. The outputs of the training data are assumed to contain Gaussian noises with distributions $\mathcal{N}(0,\ 0.03^2)$, $\mathcal{N}(0,\ 0.03^2)$, and $\mathcal{N}(0,\ 0.02^2)$, respectively. The maximum capacities of the dataset for training GP models are set to 20. Besides, the hyperparameters σ_k and l_i in the kernel are optimized by minimizing the negative log likelihood function [31], which can be returned by the GP models.

The simulation time is 20 s. The simulation results are shown in Fig. 3. Fig. 3(a) and (b) shows the tracking performance of the bicycles controlled by the proposed method and the method in [14]. Hewing *et al.* [14] developed a feedforward-PD tracking controller while the disturbance is not actively estimated and compensated for. From Fig. 3(a), one can see that the bicycle, starting from the point $(0,\ 0.5)$, gradually moves to the desired trajectory and remains extremely close to the desired trajectory thereafter. In Fig. 3(b), the tracking error, which is defined by $e_{\text{tr}} = \sqrt{(p_{xd}-p_x)^2 + (p_{yd}-p_y)^2}$, is compared. We hope that the tracking error e_{tr} could be limited to 0.1 m, so according to Theorem 1, $\alpha_3 = 0.02$ is a feasible choice. Observing the blue dashed line verifies that the designated tracking accuracy can be achieved. In Fig. 3(b), we set different values for α_3. Smaller α_3 leads to more data updates and higher tracking accuracy. That is what adjustable performance means.

Fig. 3(c) shows the information about the roll angle, the pendulum angle, and the steering angle, respectively. It can be observed that the maximum absolute values of these three angles are about 10°, 40°, and 20°, respectively. Note that the roll angle does not stabilize at zero since it is adjusted to the change of trajectory curvature.

In Fig. 3(d), the true unknown dynamics with measurement noise are compared with the estimated values. It turns out that the GP models maintain a high estimation accuracy. The estimates of D record some jumps at certain points due to the addition of new training data. Namely, the estimation errors meet the conditions in (14) and (25), and thus, the GP models are triggered to add new training data.

VI. Experiments

A. Setup

Based on the simulation results in Section V, the proposed control law with the same parameters is implemented in a real bicycle robot (see Fig. 4), whose physical parameters are the same as those in the simulation section. The steering and the pendulum are controlled by servos with position mode, so the corresponding control inputs are the real steering angle ϕ and the virtual signal η_2 in the second step of the backstepping procedure. In view that the maximum rotation rate of the servos is 45 r/min, the right-hand side of the pendulum controller (21) is modified to a saturated version with the saturation constant equal to $45 \times 2\pi \times \varepsilon/60 \approx 0.471$ rad/s. We use wireless data transmission to ensure the real-time communication between the motors and sensors of the bicycle and the host computer. The controller is implemented in MATLAB 2021a on the host computer, and the control and communication rates are 10 ms.

In this experiment, we guide the bicycle to follow different trajectories, such as a straight line and an "8"-shape trajectory. Due to the scale of indoor space, the running times of the two experiments are set as 11 and 20 s, respectively. When the bicycle is following the straight line, we setup some slopes on the ground to test the antidisturbance capability of the learning controller. Since the pendulum servo is in position mode, the uncertainty of the pendulum dynamics [the second subsystem of

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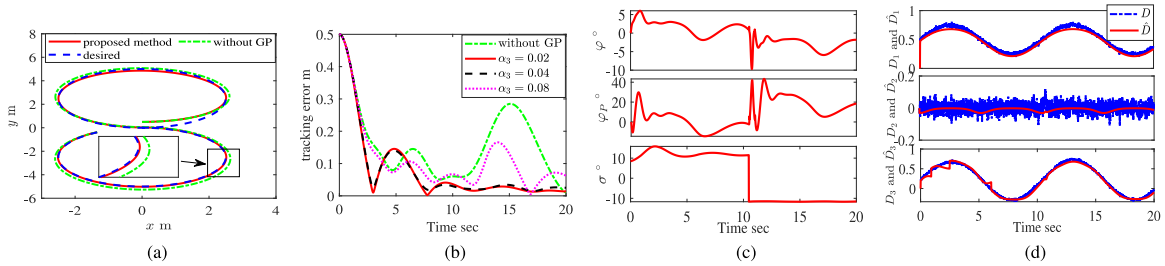
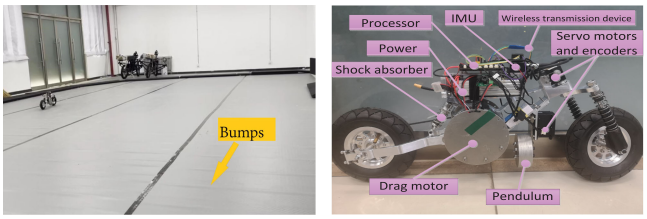


Fig. 3. Simulation results. (a) Comparison of trajectories. (b) Comparison of tracking errors. (c) Angle of roll, pendulum, and steering. (d) Performance of the GP models.



(a) (b)

Fig. 4. Pictures of the testing experiment setup. (a) Experimental environment. (b) A snapshot of the bicycle robot.

(1)] is excluded. Thus, we establish two GP models to predict the values of D_1 and D_3, which will be used in the feedback design of F_{trac} and η_2, respectively. The control parameters in these experiments are taken as $\varepsilon = 0.1,\ k_1 = k_2 = 2,\ p_0 = 1.5,\ p_1 = 0.8,\ l_1 = l_2 = 1,\ \alpha_1 = 2,$ and $\alpha_3 = 0.03$. Considering the limitations of the precision and response speed of the servo's position control, we add a PD controller with respect to the roll angle to the steering actuator to assist balance, but this is at the expense of certain tracking accuracy.

B. Results

The bicycle is initially placed near the origin of the motion-capture system. The experimental results (see Supplementary Material) are plotted in Figs. 5 and 6. We plot the real and desired trajectories of the bicycle in Figs. 5(a) and 6(a), and the corresponding root square errors in Figs. 5(b) and 6(b). It is clearly seen that the proposed controller shows smooth and favorable tracking behaviors for the two desired trajectories, with the ultimate tracking errors less than 0.1 and 0.2 m, ultimately. The downward trend of the straight-line tracking error is especially significant. Figs. 5(c) and 6(c) depict the roll angle, the pendulum angle, and the steering angle of the bicycle during the experiments. The measured rear-wheel rotation rate is converted into the longitudinal velocity and plotted in Figs. 5(d) and 6(d), from which one sees that the velocity changes continuously in a reasonable range and reaches a plateau around the desired one. In addition, we illustrate the first GP model $\hat{D}_1(x^+ \mid \mathcal{D}_\kappa)$ in Figs. 5(e) and 6(e), and the red line depicts the predicted upper bound of the estimation error for D_1. At certain time instants, events (red circles) are triggered when the predicted upper bound of the estimation error reaches the threshold, as decided in the event-triggered mechanism.

C. Discussions

In the experiment of straight-line tracking, one can observe the influence of the slope on the performance of our approach. When the vehicle moves into the slope area, the velocity decreases, which results in an upward trend of the tracking error. However, the roll angle of the vehicle does not record any large fluctuations, which verifies the antidisturbance ability of the algorithm.

In the experiment of the "8"-shape trajectory tracking, all signals including the roll, pendulum, steering angles, velocity, and tracking error remain almost steady for 20 s, except for a dramatic variation in the time interval from 10 to 13 s. This is because the direction of the normal vector of the desired trajectory shifts when $t = 10$ s, resulting in the sign change of the second derivative of the desired trajectory.

VII. Conclusion

In this article, we present a learning-based tracking control law for an underactuated bicycle robot subject to various kinds of uncertainties. By using GP to model the uncertainties, the tracking error is limited to a small range that can be explicitly given by design. Asymptotic stability of the tracking error can further be fulfilled under the situation of zero-GP modeling error. A backstepping controller is running on the roll-pendulum system to make the bicycle stabilize at the balance equilibrium manifold. Then, a comparative simulation validates the effectiveness of the control scheme. At last, we setup an experiment in which the control algorithm is embedded in a bicycle in the reduced-scale for testing.

In this article, the tracking performance of the external loop can be further improved to achieve some optimal targets by using optimization-based control. In the future, we shall attempt to endow the bicycle with the decision-making ability to complete the optimal planning/tracking by employing model predictive control or reinforcement learning. Meanwhile, we will investigate the problem of zero-speed balancing by merely using the control torque of the pendulum.

Appendix A
Proof of Theorem 1

Let us begin the proof with calculating the derivative of V_{px} alone the trajectories of the perturbed error dynamics (12)

$$\dot{V}_{px} \leqslant -\|e_x\|^2 + \frac{\partial V_{px}}{\partial e_x} B_p \left(c_\Psi M_{33}^{-1} e_{D3} + u_{rx}\right). \tag{29}$$

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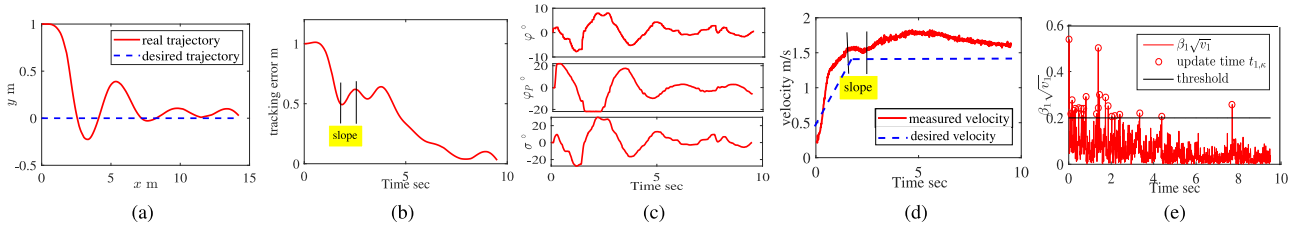

Fig. 5. Experiment results of straight line tracking. (a) Straight line trajectory. (b) Tracking error. (c) Angles of roll, pendulum, and steering. (d) Measured velocity of the bicycle. (e) Illustration of the event-triggered learning.

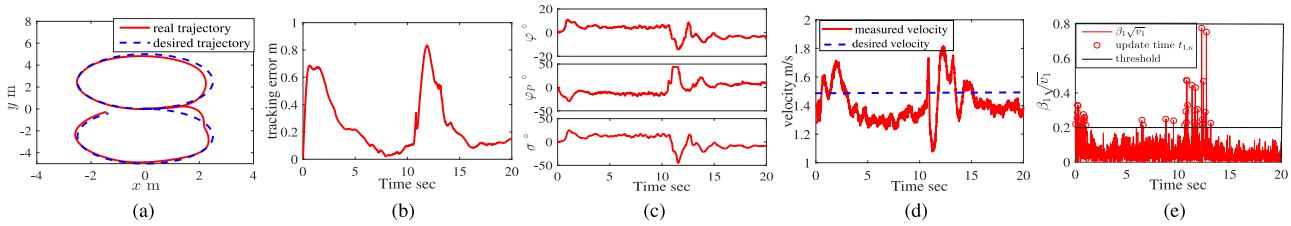

Fig. 6. Experiment results of the "8"-shape tracking. (a) "8"-shape trajectory. (b) Tracking error. (c) Angles of roll, pendulum, and steering. (d) Measured velocity of the bicycle. (e) Illustration of the event-triggered learning.

According to Lemma 1

$$\Pr\left\{\left|c_\Psi M_{33}^{-1} e_{D3}\right| \leqslant \left|M_{33}^{-1}\right| \beta_3 \sqrt{v_3\left(x|\mathcal{D}_\kappa\right)}\right\} \geqslant 1-\delta \quad \forall x \in \mathbb{X}. \tag{30}$$

Based on the data update principle (14), we conclude that $|M_{33}^{-1}|\beta_3\sqrt{v_3(x|\mathcal{D}_\kappa)} \leq \alpha_3$ holds for all $t \in [0,\ \infty)$. Set $w = (\partial V_{px}/\partial e_x)B_p \in \mathbb{R}$ and substitute (13) into (30). Then, (30) can, thus, be transformed into

$$\Pr\left\{\dot{V}_{px} \leqslant -\|e_x\|^2 + |w|\,\alpha_3 - w^2\alpha_3\right\} \geqslant 1-\delta \quad \forall x \in \mathbb{X}. \tag{31}$$

The term $|w|\alpha_3 - w^2\alpha_3$ attains a maximum value $\alpha_3/4$ at $|w| = \alpha_3/2$. Therefore, (31) can be converted into

$$\Pr\left\{\dot{V}_{px} \leqslant -\|e_x\|^2 + \alpha_3/4\right\} \geqslant 1-\delta \quad \forall x \in \mathbb{X}. \tag{32}$$

As a result, the Lyapunov candidate function V_{px} decreases $\dot{V}_{px} \leq 0$ with the probability larger than $1-\delta$ outside of the set $\mathbb{E} = \{\|e_x\| \leqslant \sqrt{\alpha_3}/2\}$. Since V_{px} also satisfies $\lambda_{\min}(P)\|e_x\|^2 \leq V_{px} \leq \lambda_{\max}(P)\|e_x\|^2$, where $\lambda_{\max}(P)$ and $\lambda_{\min}(P)$ represent the maximum and minimum eigenvalues of a symmetric positive definite matrix P, respectively, (32) allows for the conclusion that the tracking error $\|e_x\|$ converges to the ball $\mathbb{E}$ for all $x \in \mathbb{X}$. The ball $\mathbb{E}$ is independent of the initial state of e_x, then GUUB property can be, thus, derived. The same conclusion applies to e_y, and its proof is omitted here.

In addition, since $\dot{p}_{xd}$ and $\dot{p}_{xd}$ are bounded, it follows from (1) and (11) that v_{rx} , u_a, and u_Ψ are all bounded. ■

Appendix B
Proof of Theorem 2

Similar to the definition in the works of [29], the definition of estimation error for the ESO is $E_\Psi = [(u_\Psi - \Psi^{(1)})/\varepsilon^2,\ \dot{u}_\Psi - \Psi^{(2)}]^T$. Meanwhile, define $\tilde{\eta} = \eta - \eta^* \in \mathbb{R}^3$, where $\eta = [\eta_1,\ \eta_2,\ \eta_3]^T$ and $\eta^* = [\eta_1^*,\ \eta_2^*,\ \eta_3^*]^T$. The proof can be decomposed into four parts.

Part 1: Convergence of the ESO. By considering (22), we can get the dynamics of E_Ψ as

$$\varepsilon^2\dot{E}_\Psi = A_\Psi E_\Psi + \varepsilon^2 B_\Psi \ddot{u}_\Psi \triangleq g_\Psi\left(E_\Psi,\ \ddot{u}_\Psi,\ \varepsilon\right) \tag{33}$$

where $A_\Psi = \begin{bmatrix} -l_1 & 1 \\ -l_2 & 0 \end{bmatrix}$ and $B_\Psi = [0\ 1]^T$. From the expression of (33), (33) has a unique isolated root $E_\Psi = 0$ when ε is set to zero. Therefore, we find that the boundary-layer system

$$\frac{dE_\Psi}{d\tau_1} = A_\Psi E_\Psi, \quad \tau_1 = \frac{t}{\varepsilon^2} \tag{34}$$

is globally exponentially stable [23, Definition 11.1] since the polynomial $s^2 + l_1 s + l_2$ is Hurwitz.

Part 2: Convergence of η. Using (19), (21), and (23), we can derive the dynamics of η as

$$\varepsilon\dot{\eta} = g_\eta\left(z,\ \eta,\ E_\Psi,\ \varepsilon\right) = \begin{Bmatrix} -\eta_1 + \eta_1^* \\ f_1\left(\varsigma_1,\ \eta_2\right) - f_1\left(\varsigma_1,\ \eta_2^*\right) + g_1^{\Psi^{(2)}} - g_1^{\ddot{\Psi}} + \hat{D}_1 - D_1 \\ -\eta_3 + \eta_3^* \end{Bmatrix} \tag{35}$$

where $g_1^{\Psi^{(2)}} = g_1(\varphi,\ \sigma,\ \dot{\sigma},\ v_{rx},\ \Psi^{(2)})$ and $g_1^{\ddot{\Psi}} = g_1(\varphi,\ \sigma,\ v_{rx},\ \ddot{\Psi})$. Note that $f_1(\cdot)$ is a strictly decreasing function with respect to ς_3. Then, the right-hand side of (35) indicates

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that (35) has a unique isolated root $\eta = h(z)$ with $h(z) = \eta^*$ under the situation of $\varepsilon = 0$. As a result, the boundary-layer system for (35) is given by

$$\frac{d\tilde{\eta}_1}{d\tau_2} = -\tilde{\eta}_1,\ \frac{d\tilde{\eta}_2}{d\tau_2} = f_1(\varsigma_1,\ \eta_2) - f_1(\varsigma_1,\ \eta_2^*),\ \frac{d\tilde{\eta}_3}{d\tau_2} = -\tilde{\eta}_3. \tag{36}$$

By using the mean value theorem, we know that there exists a $\lambda_1 \in (0,\ 1)$, such that $f_1(\varsigma_1,\ \eta_2) - f_1(\varsigma_1,\ \eta_2^*) = \frac{\partial f_1}{\partial \varsigma_3}|_{\varsigma_3=\lambda_1\eta_2+(1-\lambda_1)\eta_2^*}\tilde{\eta}_2$. Thanks to Property 1, we know that $\frac{\partial f_1}{\partial \varsigma_3}$ is bounded and negative for all $(\varphi,\ \varphi_P) \in \mathbb{B}_\varphi \times \mathbb{B}_{\varphi P}$. Thus, it is possible to consider the Lyapunov candidate function $V_\eta = \frac{1}{2}\|\tilde{\eta}\|^2$. Calculating the scaled time derivative of the Lyapunov candidate function along the trajectory of (36) gives

$$\frac{dV_\eta}{d\tau_2} = -\tilde{\eta}_1^2 - \tilde{\eta}_3^2 + \left.\frac{\partial f_1}{\partial \varsigma_3}\right|_{\varsigma_3=\lambda_1\eta_2+(1-\lambda_1)\eta_2^*} \tilde{\eta}_2^2 \leqslant -\lambda_\eta V_\eta \tag{37}$$

in which $\lambda_\eta = 2\min\left\{1,\ \inf_{\lambda\in(0,1)}\left\{-\frac{\partial f}{\partial \varsigma_3}|_{\varsigma_3=\lambda_1\eta_2+(1-\lambda_1)\eta_2^*}\right\}\right\}$.

Based on (37), it can be easily proved that the origin of (36) is an exponentially stable equilibrium point.

Part 3: Stability condition for the reduced system. Based on the singular perturbation theory, applying $\eta = \eta^*$ and $\Psi^{(2)} = \ddot{\Psi}$ to the closed-loop error dynamics (26), we can get the reduced system of (26) as follows:

$$\begin{cases} \dot{z}_1 = -k_1 z_1 + M_{11}^{-1} z_2 \\ \dot{z}_2 = -k_2 z_2 + f_1(\varsigma_1,\ \varsigma_3) - f_1(\varsigma_1,\ \eta_2) + e_{D1} \\ \dot{z}_3 = -k_3 z_3 + z_4 \\ \dot{z}_4 = -k_4 z_4 + b_1 e_{D1} + b_2 e_{D2}. \end{cases} \tag{38}$$

For system (38), define a Lyapunov function $V_z = \frac{1}{2}\sum_{i=1}^4 z_i^2$. Computing the time derivative of V_z gives

$$\begin{aligned} \dot{V}_z &= -k_1 z_1^2 - k_2 z_2^2 - k_3 z_3^2 - k_4 z_4^2 + M_{11}^{-1} z_1 z_2 + z_3 z_4 \\ &\quad + z_4(b_1 e_{D1} + b_2 e_{D2}) + z_2(f_1(\varsigma_1,\ \varsigma_3) - f_1(\varsigma_1,\ \eta_2) + e_{D1}) \\ &\leqslant -k_1 z_1^2 - k_2 z_2^2 - k_3 z_3^2 - k_4 z_4^2 + c_1 z_1 z_2 + z_3 z_4 \\ &\quad + c_2 z_2 z_3 + z_2 e_{D1} + z_4(b_1 e_{D1} + b_2 e_{D2}) \end{aligned} \tag{39}$$

where c_1 and c_2 are given in Property 3. From the relationship $z_i z_{i+1} \leq \frac{1}{2}(z_i^2 + z_{i+1}^2)$ and Lemma 1, we further get that V_z satisfies the following inequality in probabilistic form:

$$\begin{aligned} \Pr\Bigg\{\dot{V}_z \leqslant &-\left(k_1 - \frac{c_1}{2}\right) z_1^2 - \left(k_2 - \frac{1+c_1+c_2}{2}\right) \\ &z_2^2 - \left(k_3 - \frac{1+c_2}{2}\right) z_3^2 - (k_4 - 1) \\ &z_4^2 + \frac{1}{2}\left(|e_{D1}|^2 + |b_1 e_{D1} + b_2 e_{D2}|^2\right)\Bigg\} \geqslant 1 - \delta. \end{aligned} \tag{40}$$

Notice that $\frac{1}{2}(|e_{D1}|^2 + |b_1 e_{D1} + b_2 e_{D2}|^2) \leq \frac{\alpha_1^2\varepsilon^2 + \alpha_2^2}{2}$ holds due to the event-triggered mechanism in (25). At $\varepsilon = 0$, (40) further implies

$$\Pr\left\{\dot{V}_z \leqslant -\lambda_z V_z + \frac{\alpha_2^2}{2}\right\} \geqslant 1 - \delta \tag{41}$$

where $\lambda_z = 2\min\{k_1 - \frac{c_1}{2}, k_2 - \frac{1+c_1+c_2}{2}, k_3 - \frac{1+c_2}{2}, k_4 - 1\}$. Solving (41) yields

$$\Pr\left\{V_z(t) \leqslant \frac{\alpha_2}{2\lambda_z} + \left(V_z(0) - \frac{\alpha_2}{2\lambda_z}\right)\exp(-\lambda_z t)\right\} \geqslant 1 - \delta. \tag{42}$$

Thus, we can make the conclusion that if the backstepping controller gains are set to satisfy $k_1 > c_1/2,\ k_2 > (1 + c_1 + c_2)/2$, and $k_3 > (1 + c_2)/2, k_4 > 1$, the reduced system of the closed-loop error dynamics is UUB.

Part 4: Singular perturbation analysis. It follows from [32] that the conditions for applying [23, Theorem 11.1] can be verified. Then, we can conclude that there exists a small positive constant $\varepsilon_1 > 0$ such that for any $\varepsilon \in (0,\ \varepsilon_1)$, the singular perturbation problem of (26), (33), and (35) has a unique solution $z(t,\ \varepsilon),\ E_\Psi(t,\ \varepsilon)$, and $\eta(t,\ \varepsilon)$ in $t \in [0,\ \infty)$, and

$$\begin{cases} z(t,\ \varepsilon) - \bar{z}(t) = O(\varepsilon) \\ E_\Psi(t,\ \varepsilon) - \bar{E}_\Psi(t) = O(\varepsilon^2) \\ \tilde{\eta}(t,\ \varepsilon) - \bar{\eta}(t) - h(\bar{z}(t)) = O(\varepsilon) \end{cases} \tag{43}$$

holds uniformly for $t \in [0,\ \infty)$, where $\bar{z}(t),\ \bar{E}_\Psi(t)$, and $\bar{\eta}(t)$ are the solutions of the reduced system (38) and boundary-layer systems (34) and (36), satisfying

$$\begin{cases} \Pr\left\{\lim_{t\to\infty}\|\bar{z}(t)\| \leqslant \frac{\alpha_2}{\sqrt{\lambda_z}}\right\} \geqslant 1 - \delta \\ \bar{E}_\Psi(t) = \exp\left(A_\Psi t/\varepsilon^2\right) E_\Psi(0) \\ \|\bar{\eta}(t)\| = \exp(-\lambda_\eta t/\varepsilon)\|\tilde{\eta}(0)\| \end{cases} \tag{44}$$

according to the proofs in Parts 1–3. Therefore, we immediately obtain (28) with a positive constant N_ε, which means that all the signals in the closed-loop system (17) are bounded (by probability). Furthermore, from (42) and (28), we can find that $z(t)$ converges to the bound $\mathbb{B}_z = \{\|z(t)\| \leq \alpha_2/\sqrt{\lambda_z} + N_\varepsilon\varepsilon\}$ with exponential decay, and the bound $\mathbb{B}_z$ can be reduced by properly raising the gains k_i or making α_2 and ε smaller. This completes the proof of Theorem 2. ■

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